1.11.1

(b)

p	q	p∨q	p<->q	(p∨q)^(p<->q)	(p∨q)^(p<->q)=>p
T	T	T	T	T	T
T	F	T	F	F	T
F	T	T	F	F	T
F	F	F	T	F	T

(j)

p	q	p→q	q→p	p	¬(p→p)
T	T	T	T	T	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	T	F	F

1.11.2

(b)

p	q	p→q
T	T	T
T	F	F
F	T	T
F	F	T

Since it is given that p→q is True, pick the rows where the value of p→q is True. This result in the Following Table:

p	q	p→q

T	T	T
F	T	T
F	F	T

Also given that $\neg p$ is true, or p is False, pick the rows where the value of p is False. The results in the Following table.

p	q	$p \to q$
F	T	T
F	F	T

1.11.3

(b) He studied For the $\underline{\text{Test}}_{p}$ or he $\underline{\text{failed The test}}_{\neg p}$ or both

	$p \vee \neg p$	
p	$\neg p$	$p \vee \neg q$
T	F	T
F	T	T

Q: He passed The test

$(p \vee \neg p) \wedge Q \to$

He studied for the test or he failed the test
or he passed the test

1.12.3

(b) 1. $P \to (q \wedge r)$	Hypothesis	
2. $\neg q$	Hypothesis	
3. $\neg p \vee (q \wedge r)$	Conditional disjunction, 1	
4. $\neg p \vee (F \wedge r)$	Substitute 2 in 3	
5. $\neg p \vee F$	Universal Bound, 4	
6. $\neg p$	Identity . 5.	

(d) 1. $(p \vee q) \to r$	Hypothesis	
2. $\neg r$	Hypothesis	
3. q	Hypothesis	
4. $\neg (p \wedge q)$	modus Tollens 1, 2	
5. $\neg (p \vee \neg q)$	De Morgan's law, 4	
6. $\neg p$	Disjunctive Syllogism 3, 5.	

1.12.3

(b) 1. $p \to q$	Premise	
2. p	Premise	
3. $(p \to q) \wedge p$	Conjunction (1, 2)	
4. $(\neg p \vee q) \wedge p$	Conditional disjunction (3)	
5. $(\neg p \wedge p) \vee (q \wedge p)$	Distributive law (4)	
6. $F \vee (q \wedge p)$	Negation (5)	
7. $(q \wedge p)$	Identity (6)	

8.9 Specialization (7)

2.1.1

(b) $n = -101$

$n = -101$ is odd

$n = 2(-101) + 1$

$n = 2.(-100)$

2.1.2

(d) $-2n^2 - 5$

2.1.3

(b) $n = -\frac{5}{1}$

Rational

2.1.4

(d) $n = 75$

1, 3, 5, 15, 25, 75

2.1.5

(d) Prime

(F) Composite: $1 < 2 < 56328$

2.1.6

(d) $x < 7$

2.2.1

(b) In such a proof, we either explicitly find an example x in the domain D for which the propositional function $p(x)$ is true, or we give a set of directions for finding such an x.

2.2.2

(b) Case 1: $n = 0$. Then $2^{(n+2)} = 4$ is clearly greater than 1 (i.e. $3^n = 3^0 = 1$)

Case 2: $n = 1$. Then $2^{(n+2)} = 8$ is clearly greater than 3 (i.e. $3^1 = 3$).

Case 3: $n = 2$. Then $2^{(n+2)} = 16$ is clearly greater than 9 (i.e. $3^n = 3^2 = 9$)

Case 4: $n = 3$. Then $2^{(n+2)} = 32$ is clearly greater than 27 (i.e. $3^n = 3^3 = 27$).

2. 2. 3

(d) 3 is one possible Counterexample. 3 can not be expressed as the sum of the Square of two positive integers.

2. 2.4

(b) Use proof by Contraction to show that there is no greatest integer. there is no object with preperty begin by supposing the negation: that there is an object with the property.

2.2.5

(F) yes. The number is -1.

2.3.1

(d) $n^2 + (n+1)^2$

$n^2 + n^2 + 2n + 1$

$2n^2 + 2n + 1$

The first part of this is even because it is divisible by 2. By adding 1. it must become odd.

2.3.2

(b) Taking m as an interger such as $x2 = m \cdot w_4$ is invalid.

x and 2 should be multiplied first.

2.3.3

(b) Here we considering the general form of an odd interger: it is actually $n = 2k+1$ for some interger k and $m = 2j+1$ for some interger

But in the proof we are assuming only $n = 2k+1$ and $m = 2j+1$ which is wrong.

2.4.1

(d) The product of two odd numbers is an odd numbers.
Let m and k be any intergers. This means that $2m+1$ and $2k+1$ are odd numbers.

2.4.4

(d) $x^2 > y^2$

$xx > yy$

$\sqrt{xx} > \sqrt{yy}$

$x > y$

(h) $(2m+1 + 2n+1)/2$

$= (2m+2n+2)/2$

$= m+n+1$

2.5.1

(d) $n+1 = (2k+1)^2 - 2(2k+1) + 7$

$\rightarrow (2k+1) = (4k^2 + 4k + 1) - 4k - 2 + 7 - 1$

$\rightarrow 2k = 4k^2 + 4$

$\rightarrow 2(2k^2 - k + 2)$.

2.5.3

(b) Primes greater than 3 can't be 2.
Since they are primes, they can't be multiples of 3 and can't be even.

2.6.4

(b) Proof by Contradiction.

2.6.5

(b) $n^2 = (2x-1)^2$

2.6.6

(b)